

1. Core Predictive Models

The project relies on two primary regression approaches to understand what drives customer satisfaction.

- **Multiple Linear Regression:** This model was implemented by treating the customer satisfaction score as a continuous numerical variable. It allows to estimate a straight-line relationship between the predictors (like total spending or weather) and satisfaction, quantifying the direction and magnitude of their influence.
- **Ordinal Logistic Regression:** This model was used to recognize that the satisfaction ratings (3, 4, and 5) are actually ordered categories rather than purely continuous numbers. It provides a more mathematically rigorous way to evaluate ordered categorical outcome data, and comparing it with the linear model helps evaluate the respective strengths and interpretability of both approaches.

2. Formal Hypothesis Tests

To determine if the relationships observed in the data were statistically significant (at a 5% significance level, or $\alpha=0.05$), the report outlines several specific tests derived from the regression models.

- **F-test (Linear) / Likelihood Ratio Test (Ordinal):** To evaluate the *overall model significance*, this tests the null hypothesis that there is no relationship between customer satisfaction and *any* of the combined predictor variables.
- **t-test (Linear) / Wald test (Ordinal):** To evaluate *individual predictor significance*. Specifically, this was used to test the isolated effects of variables like total spending and special offers. For example, it determines if transactions with special offers have a statistically different average satisfaction than those without.
- **Partial F-test:** To evaluate *differences across multi-level categorical variables*, specifically Item Types (Sandwich, Drink, Salad, Burger) and Weather Conditions (Sunny, Cloudy, Rainy). It tests whether at least one category within these variables has a significantly different satisfaction score compared to the others.

3. Model Diagnostic Tests

To ensure the findings from the multiple linear regression were reliable and valid, several diagnostic tests (visual and mathematical) were performed to check the underlying assumptions of the model.

- **Residuals vs Fitted Plot (Testing Linearity):** To verify the assumption of linearity. The plot showed a relatively random scatter of points around a horizontal line with no clear pattern, confirming that the model adequately captured the systematic patterns in the data without missing non-linear relationships.
- **Q-Q Residuals Plot (Testing Normality):** To verify that the residuals (the errors of the model) are approximately normally distributed. This assumption is required to ensure that the hypothesis tests (like the t-tests and F-tests) and their confidence intervals are mathematically valid.
- **Scale-Location Plot (Testing Homoscedasticity):** To check for constant variance in the residuals across different fitted values. A flat line with random scatter confirmed homoscedasticity, meaning the model's predictive accuracy is consistent regardless of whether it is predicting a high or low satisfaction level.
- **Cook's Distance & Residuals vs Leverage Plots (Testing for Outliers/Influence):** To identify any specific data points that were heavily skewing the results. The tests confirmed that no single observation had an excessive influence (Cook's distance < 1) that would distort the regression coefficients, validating the model's robustness.
- **Variance Inflation Factor / VIF (Testing Multicollinearity):** Calculated using the car package in R to ensure there was an absence of multicollinearity. This checks that the predictor variables are not too highly correlated with each other, which would make it difficult to determine their individual impacts.